

Machine Learning End-to-End Roadmap

Module 1: Foundations

Learn Python, NumPy, Pandas, statistics, probability, and linear algebra.

Key concepts: vectors, matrices, derivatives, probability distributions, Bayes theorem.

Focus on intuition and examples.

Module 2: Supervised Learning

Algorithms: Linear Regression, Logistic Regression, KNN, Decision Trees, Random Forest, SVM, Naive Bayes.

Each includes intuition, math, use cases, advantages, disadvantages, and Python implementation.

Module 3: Unsupervised Learning

Algorithms: K-Means, Hierarchical Clustering, DBSCAN, PCA.

Focus on clustering and dimensionality reduction.

Module 4: Model Evaluation

Regression: MSE, RMSE, MAE, R^2 .

Classification: Accuracy, Precision, Recall, F1 Score, ROC-AUC.

Understand confusion matrix and metric selection.

Module 5: Feature Engineering

Handling missing values, encoding, scaling, and feature selection techniques.

Module 6: Model Improvement

Overfitting vs underfitting, cross-validation, hyperparameter tuning.

Module 7: Projects

Beginner: House price prediction.

Intermediate: Customer churn, spam detection.

Advanced: Recommendation system, fraud detection.

Module 8: End-to-End Pipeline

Steps: Data collection → cleaning → EDA → modeling → evaluation → deployment.

Module 9: Deployment

Deploy using Flask/FastAPI, Docker basics, APIs.

Module 10: Interview Preparation

Common ML questions, algorithm comparisons, real-world scenarios.